

On-Device Speech Scoreboard

An Android app that writes down what was said in a conversation and who said it, running entirely on the phone with no network connection. This is the current pipeline — what it runs, how it is scheduled, and what it measures on two tablets in English and German.

ElevenLabs voices

Piper & macOS voices, long corpus

Lenovo TB336FU · MediaTek MT8755

Xiaomi 25097RP43I · Snapdragon 8s Gen 4

2026-08-25 to 08-28

HOW A RECORDING BECOMES A LABELLED TRANSCRIPT

Audio is captured at 16 kHz mono. Five stages run over it, and the results exercise all of them at once.

1. A segmentation model cuts the audio into turns, meaning stretches where one person is talking. It treats silence as the cue for where a turn ends.
2. An embedding model converts each turn into a voiceprint: a vector describing the voice itself rather than the words.
3. The voiceprints are clustered. With no speaker count given, clustering cuts at a 0.5 similarity threshold; small fragments are then folded into larger clusters.
4. Each cluster is compared against the voiceprints of people who enrolled and takes the closest name. Clusters with no close match stay unnamed.
5. A recogniser transcribes the audio, and its word timings are matched to the turn ranges, so every word ends up underneath a speaker.

How it is scheduled (since 28 Aug). Stages 1–4 and stage 5 are data-independent, so the two branches run concurrently and the wall clock is the longer branch. Long recordings split into ~5-minute chunks diarised on parallel lanes, one per budgeted core; each chunk is folded and named while later chunks are still being diarised; transcription decodes on two recogniser instances taking alternate slices; and every model stays warm between runs, so a repeat run does zero model loads.

Cutting silence is optional: silent parts are removed before attribution (the recogniser still gets everything). Off for the synthetic long files below, which have no silence to cut; on a real meeting it removed 36.7%.

Coverage	WER
How much of the expected text the transcript reached. Read before the error rate: below 90% the transcript stopped early and its error rate means nothing.	Word error rate. Share of words the recogniser got wrong, numerals spelled out so "14001" and "fourteen thousand and one" agree. Lower is better.

THE CURRENT PIPELINE'S MODELS All local through sherpa-onnx, 165 MB together, on the CPU via XNNPACK — its ARM int8 kernels, now the default provider.

STAGE	MODEL	SIZE
Turn segmentation	pyannote segmentation 3.0 sherpa-onnx-pyannote-segmentation-3-0	5.7 MB
Voiceprints	3D-Speaker CAM++ 3dspeaker_speech_camplusplus_sv_zh_en_16k-common_advanced	27.0 MB
Transcription	NeMo FastConformer en/de/es/fr, int8, per-word timings sherpa-onnx-nemo-fast-conformer-transducer-en-de-es-fr-14288-int8	132 MB

A heavier shelf stays available for the one case this set gives up: NVIDIA Parakeet 0.6B (639 MB, ~1.3 WER points better on dense English, level on German at 2.47% vs 2.5%) and ERes2Net-base (37.8 MB, separates same-gender voices CAM++ merges). The eleven_de rows below show that trade in numbers.

THE VOICES

Synthetic conversations with per-sample ground truth: who spoke, from when, to when. The ~20-minute files are rendered locally — Piper (Kerstin f, Thorsten m, German) and macOS voices (Daniel m, Samantha f, English); rougher renderings than a premium engine, which is why their WER reads high. eleven_de is ElevenLabs (George m, Daniel m), close enough to real speech that telling two men apart is a fair test.

The pairing is the difficulty. Two male voiceprints sit at 0.734 cosine similarity against 0.91 for two recordings of the same person — the ceiling every later stage inherits, because once a cluster holds both voices only one name can attach to it.

RESULTS ON THE CURRENT PIPELINE Warm models, VAD off, re-measured 28 August from the devices' own databases. "Speakers" and "words" are the two concurrent branches; "whole run" adds file reading and scoring on top of the longer one. Speaker accuracy is by word.

FILE	VOICES	LG	DEVICE	AUDIO	SPEAKERS	WORDS	WHOLE RUN	REALTIME	WER	SPEAKER
german_mixed_long	one woman, one man · Piper	DE	Xiaomi	22m 02s	48.5s	40.9s	49.9s	26.5×	10.3%	97.9%
german_mixed_long	one woman, one man · Piper	DE	Lenovo	22m 02s	148.3s	102.8s	150.7s	8.8×	10.3%	97.9%
long_en	one man, one woman · macOS voices	EN	Xiaomi	20m 36s	47.2s	36.4s	58.5s	21.1×	2.65%	95.6%
long_en	one man, one woman · macOS voices	EN	Lenovo	20m 36s	146.3s	101.6s	169.6s	7.3×	2.65%	95.6%
eleven_de	two men · the CAM++ trade, see note	DE	Xiaomi	3m 05s	7.6s	6.5s	8.6s	21.5×	2.47%	59.2%
eleven_de	two men · the CAM++ trade, see note	DE	Lenovo	3m 05s	22.4s	13.8s	22.5s	8.2×	2.47%	59.2%

What changed to get here, 27–28 August

The objective was a 20-minute recording in under 60 seconds; the Xiaomi started at ~115. In order: the FastConformer swap and the XNNPACK default (115→64 s); a recogniser-aware thread budget and two diarise lanes (→55.5); then the 28 August scheduling — chunks folded and named while later chunks still diarise, one single-thread lane per budgeted core (four on the Xiaomi), a second recogniser instance on alternate slices, and warm models with zero loads on a repeat run (→47–50 s). The Lenovo went 247.9→150.7 s on the German file, –39%. Every step produced byte-identical output; WER and speaker accuracy matched digit-for-digit throughout.

Both tablets give the same numbers

Attribution is deterministic given the same audio and models, and it holds across chips: every file in the table scores digit-for-digit the same on both tablets — WER 10.3239% and speaker accuracy 97.9140% on the German file on both the Snapdragon and the MediaTek. Speed is the only thing the silicon changes, by roughly 3× between these two.

Two scheduling findings

sherpa runs one window or slice at a time, so parallel *instances* beat more threads: on the MediaTek, two one-thread recogniser instances beat one two-thread instance by 28% — per-operation barriers were wasting the second thread. And warmth is worth real time on entry silicon: the Lenovo's first run after an install took 177.7 s cold, the repeat 148.7 s with zero model loads. VAD was off for the synthetic long files — wall-to-wall speech gives it nothing to cut, and its pass alone cost the Lenovo ~40 s; it stays on for real recordings.

The CAM++ trade, and the ceiling that holds

The eleven_de rows above are the trade made visible: CAM++ names one of the two men and scores **59.2%** where ERes2Net, on the same file, keeps them apart at 98.2%. Enrolled from their own voices the two men scored 0.932 and 0.929 against one cluster, a gap the naming guard refuses to guess on. A mixed pair separates under either model — the long German file holds 97.9% under CAM++ — so the current set is the right default except when two same-gender voices must be told apart; then the heavier shelf earns its 639 MB.

Sources: docs/diarization-benchmark.md, docs/data/README.md. Scored with diarscore.py and wer.py. Re-scored from the devices' own databases, 2026-08-28. Model identifiers and sizes from the app's download catalogue.